

Tensor Product: D-F 10.4

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Let R, S be two rings with $1 \neq 0$.
Further, R is a subring of S , and $1_R = 1_S$.

We have

1. $0_R = 0_S$
2. $1_R = 1_S$
3. $R \subseteq S$

1 Construction of the Tensor Product $S \otimes_R N$

Let $(R, N, +_N, \cdot_N)$ be a left R -module.

1.1 Step 1: The free $\mathbb{Z}$ -module on $S \times N$

Let $\mathcal{F} := F_{\mathbb{Z}}(S \times N)$ be the free $\mathbb{Z}$ -module on the set $S \times N$, with inclusion map $\iota : S \times N \rightarrow \mathcal{F}$

For each $(s, n) \in S \times N$, note $\iota((s, n)) \in \mathcal{F}$ (we write, for simplicity, $\iota(s, n)$).

As usual, we suppress the ι -notation and write (s, n) for this basis element.

Thus every element of $\mathcal{F}$ has a unique expression

$$\sum_{i=1}^m z_i(s_i, n_i), \quad z_i \in \mathbb{Z},$$

1.2 Step 2: The subgroup of relations

Define the following subsets of $\mathcal{F}$:

$$A_1 := \{f \in \mathcal{F} \mid \exists(s_1, s_2 \in S) \exists(n \in N) f = \iota(s_1 +_S s_2, n) -_{\mathcal{F}} \iota(s_1, n) -_{\mathcal{F}} \iota(s_2, n)\}$$

$$A_2 := \{f \in \mathcal{F} \mid \exists(s \in S) \exists(n_1, n_2 \in N) f = \iota(s, n_1 +_N n_2) -_{\mathcal{F}} \iota(s, n_1) -_{\mathcal{F}} \iota(s, n_2)\}$$

$$A_3 := \{f \in \mathcal{F} \mid \exists(s \in S) \exists(r \in R) \exists(n \in N) f = \iota(s \cdot_S r, n) -_{\mathcal{F}} \iota(s, r \cdot_N n)\}$$

$$H := \langle A_1 \cup A_2 \cup A_3 \rangle \leq (\mathcal{F}, +_{\mathcal{F}})$$

Since $\mathcal{F}$ is a $\mathbb{Z}$ -module, H is also a $\mathbb{Z}$ -submodule of $\mathcal{F}$. So, $H = R(A_1 \cup A_2 \cup A_3)$

1.3 Step 3: Definition of $S \otimes_R N$

Define $S \otimes_R N := \mathcal{F}/H$. In its full $\mathbb{Z}$ -module form, we have

$$(\mathbb{Z}, \mathcal{F}/H, +_2, \cdot_2)$$

For $s \in S$ and $n \in N$, denote the coset of (s, n) in $\mathcal{F}/H$ by

$$s \otimes n := \iota(s, n) +_{\mathcal{F}} H.$$

These elements are called **simple tensors**.

1.4 Step 4: The defining relations in the quotient

Lemma 1.1. (*basic identities of $S \otimes_R N$*)

$\forall s, s_1, s_2 \in S, \forall r \in R, \text{ and } \forall n, n_1, n_2 \in N, \text{ we have}$

1. $(s_1 +_S s_2) \otimes n = s_1 \otimes n +_2 s_2 \otimes n$
2. $s \otimes (n_1 +_N n_2) = s \otimes n_1 +_2 s \otimes n_2$
3. $(s \cdot_S r) \otimes n = s \otimes (r \cdot_N n)$

Proof. **First**, by definition of A_1 , the element $\iota((s_1 +_S s_2), n) -_{\mathcal{F}} \iota((s_1), n) -_{\mathcal{F}} \iota((s_2), n) \in A_1 \subseteq H$

$$\implies \iota((s_1 +_S s_2), n) +_{\mathcal{F}} H = (\iota((s_1), n) +_{\mathcal{F}} \iota((s_2), n)) +_{\mathcal{F}} H.$$

Passing to quotient notation, this says

$$(s_1 +_S s_2) \otimes n = (\iota((s_1), n) +_{\mathcal{F}} H) +_2 (\iota((s_2), n) +_{\mathcal{F}} H) = s_1 \otimes n +_2 s_2 \otimes n.$$

Second, by definition of A_2 , the element $\iota((s, n_1 +_N n_2)) -_{\mathcal{F}} \iota((s, n_1)) -_{\mathcal{F}} \iota((s, n_2)) \in A_2 \subseteq H$.

$$\implies \iota((s, n_1 +_N n_2)) +_2 H = (\iota((s, n_1)) +_{\mathcal{F}} \iota((s, n_2))) +_2 H.$$

$$\implies s \otimes (n_1 +_N n_2) = s \otimes n_1 +_2 s \otimes n_2.$$

Third, by definition of A_3 , the element $\iota((s \cdot_S r), n) -_{\mathcal{F}} \iota((s, r \cdot_N n)) \in A_3 \subseteq H$

$$\implies \iota((s \cdot_S r), n) +_2 H = \iota((s, r \cdot_N n)) +_2 H$$

$$\implies (s \cdot_S r) \otimes n = s \otimes (r \cdot_N n).$$

□

1.5 Step 5: Some immediate additive consequences

Lemma 1.2. *For all $n \in N$ and $s \in S$,*

$$0_S \otimes n = 0_{S \otimes_R N} = s \otimes 0_N$$

Proof. Using the first defining relation with $s_1 = s_2 = 0_S$, we obtain

$$0_S \otimes n = (0_S +_S 0_S) \otimes n = 0_S \otimes n +_2 0_S \otimes n \implies 0_{S \otimes_R N} = 0_S \otimes n$$

Similarly, using the second defining relation with $n_1 = n_2 = 0_N$,

$$s \otimes 0_N = s \otimes (0_N + 0_N) = s \otimes 0_N +_2 s \otimes 0_N \implies 0_{S \otimes_R N} = s \otimes 0_N$$

□

Lemma 1.3. *For all $s \in S$ and $n \in N$,*

$$(-s) \otimes n = -(s \otimes n)_{S \otimes_R N} = s \otimes (-n)$$

Proof. Observe

$$0_S \otimes n = (s +_S (-s)) \otimes n = s \otimes n +_2 (-s) \otimes n \implies -(s \otimes n)_{S \otimes_R N} = (-s) \otimes n$$

$$s \otimes 0_N = s \otimes (n +_N (-n)) = s \otimes n +_2 s \otimes (-n) \implies -(s \otimes n)_{S \otimes_R N} = s \otimes (-n)$$

□

Now, take any $x \in S \otimes_R N$, $\exists f \in \mathcal{F}$ s.t. $x = f +_{\mathcal{F}} H$.

$\implies \exists! c_f : \iota(S \times N) \rightarrow \mathbb{Z}$ s.t.

$$\begin{aligned} f &= \sum_{\iota(s,n) \in \text{supp}(c_f)} c_f(\iota(s,n)) \cdot_{\mathcal{F}} \iota(s,n) = \sum_{finite} \pm \iota(s,n) \\ \implies x &= f +_{\mathcal{F}} H = \left(\sum_{finite} \pm \iota(s,n) \right) +_{\mathcal{F}} H = \sum_{finite} s \otimes n \end{aligned}$$

Such an expression is generally not unique.

1.6 Step 6: Definition of the left S -action

For each fixed $s' \in S$, define a set map $\theta_{s'} : S \times N \rightarrow \mathcal{F}$ as,

$$(s, n) \mapsto \iota(s' \cdot_S s, n)$$

By universal property, $\exists! T_{s'} \in \text{Hom}_{\mathbb{Z}}(\mathcal{F}, \mathcal{F})$ s.t. $\theta_{s'} = T_{s'} \circ \iota$. For illustration, we have

$$T_{s'} \left(\sum_{i=1}^m z_i \cdot_{\mathcal{F}} \iota(s_i, n_i) \right) = \sum_{i=1}^m z_i \cdot_{\mathcal{F}} \iota(s' \cdot_S s, n)$$

Lemma 1.4. For every $s' \in S$,

$$T_{s'}(H) \subseteq H$$

Proof. **S.T.S.** $T_{s'}$ sends $A_1 \cup A_2 \cup A_3$ into itself.

First, let $g = \iota(s_1 +_S s_2, n) -_{\mathcal{F}} (s_1, n) -_{\mathcal{F}} (s_2, n) \in A_1$. We have

$$T_{s'}(g) = \iota(s' \cdot_S (s_1 +_S s_2), n) -_{\mathcal{F}} \iota(s' \cdot_S s_1, n) -_{\mathcal{F}} \iota(s' \cdot_S s_2, n) = \iota((s' \cdot_S s_1 +_S s' \cdot_S s_2), n) -_{\mathcal{F}} \iota(s' \cdot_S s_1, n) -_{\mathcal{F}} \iota(s' \cdot_S s_2, n)$$

Second, let $g = \iota(s, n_1 +_N n_2) -_{\mathcal{F}} \iota(s, n_1) -_{\mathcal{F}} \iota(s, n_2) \in A_2$, we have

$$T_{s'}(g) = \iota(s' \cdot_S s, n_1 +_N n_2) -_{\mathcal{F}} \iota(s' \cdot_S s, n_1) -_{\mathcal{F}} \iota(s' \cdot_S s, n_2) \in A_2$$

Third, let $g = (sr, n) - (s, rn) \in A_3$, we have

$$T_{s'}(g) = \iota(s' \cdot_S (s \cdot_S r), n) -_{\mathcal{F}} \iota(s' \cdot_S s, r \cdot_N n) = \iota((s' \cdot_S s) \cdot_S r, n) -_{\mathcal{F}} \iota(s' \cdot_S s, r \cdot_N n) \in A_3$$

□

Let $\pi : \mathcal{F} \rightarrow \mathcal{F}/H$ be the natural projection.

Then, $\pi \circ T_{s'} \in \text{Hom}_{\mathbb{Z}}(\mathcal{F}, \mathcal{F}/H)$ and $\forall h \in H$, we have

$$(\pi \circ T_{s'})(h) = 0_{\mathcal{F}/H}$$

Thus, $H \subseteq \ker(\pi \circ T_{s'})$.

Now, we factor through by H and obtain $\overline{T_{s'}} \in \text{Hom}_{\mathbb{Z}}(\mathcal{F}/H, \mathcal{F}/H)$,

$$f +_{\mathcal{F}} H \mapsto \pi(T_{s'}(f)) = T_{s'}(f) +_{\mathcal{F}} H$$

For illustration, we have

$$\overline{T_{s'}} \left(\sum_{i=1}^m s_i \otimes n_i \right) = \sum_{i=1}^m \overline{T_{s'}}(s_i \otimes n_i) = \sum_{i=1}^m \iota(s' \cdot_S s_i, n_i) +_{\mathcal{F}} H = \sum_{i=1}^m (s' \cdot_S s_i) \otimes n_i$$

Define $\diamond : S \times S \otimes_R N \rightarrow S \otimes_R N$ as,

$$(s', x) \mapsto \overline{T_{s'}}(x)$$

1.7 Step 7: Verification of the left S -module axioms

Theorem 1.1. *With the action defined above, $S \otimes_R N$ is a left S -module.*

Proof. **(Identity)** Let any $x \in S \otimes_R N$. We have

$$1_S \diamond x = \overline{T_{1_S}}(x) = x$$

(Distributivity) Take any $s_1, s_2 \in S$ and $x_1, x_2 \in S \otimes_R N$, where $\forall j \in \{1, 2\}$ we have

$$x_j = \sum_{i=1}^{m_j} s_i^{(j)} \otimes n_i^{(j)}$$

$$(s_1 +_S s_2) \diamond x_1 = \overline{T_{(s_1 +_S s_2)}} \left(\sum_{i=1}^m s_i^{(1)} \otimes n_i^{(1)} \right) \quad (1)$$

$$= \sum_{i=1}^m ((s_1 +_S s_2) \cdot_S s_i^{(1)}) \otimes n_i^{(1)} \quad (2)$$

$$= \sum_{i=1}^m (s_1 \cdot_S s_i^{(1)}) \otimes n_i^{(1)} +_2 (s_2 \cdot_S s_i^{(1)}) \otimes n_i^{(1)} \quad (3)$$

$$= s_1 \diamond x_1 +_2 s_2 \diamond x_1 \quad (4)$$

$$s_1 \diamond (x_1 +_2 x_2) = \overline{T_{s_1}}(x_1 +_2 x_2) = \overline{T_{s_1}}(x_1) +_2 \overline{T_{s_1}}(x_2) = s_1 \diamond x_1 +_2 s_1 \diamond x_2$$

(Associativity) Take $s_1, s_2 \in S$ and $x \in S \otimes_R N$, we have

$$(s_1 \cdot_S s_2) \diamond x = \overline{T_{(s_1 \cdot_S s_2)}} \left(\sum_{i=1}^m s_i \otimes n_i \right) \quad (5)$$

$$= \sum_{i=1}^m ((s_1 \cdot_S s_2) \cdot_S s_i) \otimes n_i \quad (6)$$

$$= \sum_{i=1}^m (s_1 \cdot_S (s_2 \cdot_S s_i)) \otimes n_i \quad (7)$$

$$= s_1 \diamond \sum_{i=1}^m (s_2 \cdot_S s_i) \otimes n_i \quad (8)$$

$$= s_1 \diamond (s_2 \diamond x) \quad (9)$$

□

We have now S -module $(S, S \otimes_R N, +_2, \diamond)$.

1.8 Step 8: Universal Property

Corollary 1.1.1. (*corollary 9*)

Assume above $\iota : N \rightarrow S \otimes_R N$. We construct

$$\Delta := \{E \in \mathcal{P}(N) \mid R\text{-submodule}(E) \wedge \exists \varphi \exists L (Injective(\varphi) \wedge \varphi \in Hom_R(N/E, L))\}$$

Then, we have

1. $\ker \iota \in \Delta$
2. $\forall E \in \Delta, \ker \iota \subseteq E$

2 Construction of General Tensor Product $M \otimes_R N$

Let R be a ring with $1 \neq 0$.

Let $(M, +_M, \cdot_M)$ be a right R -module, and let $(N, +_N, \cdot_N)$ be a left R -module.

2.1 Step 1: The free $\mathbb{Z}$ -module on $M \times N$

Let $\mathcal{F} := F_{\mathbb{Z}}(M \times N)$ be the free $\mathbb{Z}$ -module on the set $M \times N$, with equipped inclusion map $\iota : M \times N \rightarrow \mathcal{F}$.

Thus, every element $f \in \mathcal{F}$ admits a unique expression of the form

$$f = \sum_{i=1}^k z_i \cdot_{\mathcal{F}} \iota(m_i, n_i)$$

2.2 Step 2: The subgroup of relations

We now define three subsets of F corresponding to the three relations that we want to impose in the tensor product.

$$A_1 := \{ f \in \mathcal{F} \mid \exists m_1, m_2 \in M \exists n \in N \text{ such that } f = \iota(m_1 +_M m_2, n) -_{\mathcal{F}} \iota(m_1, n) -_{\mathcal{F}} \iota(m_2, n) \}.$$

$$A_2 := \{ f \in \mathcal{F} \mid \exists m \in M \exists n_1, n_2 \in N \text{ such that } f = \iota(m, n_1 +_N n_2) -_{\mathcal{F}} \iota(m, n_1) -_{\mathcal{F}} \iota(m, n_2) \}.$$

$$A_3 := \{ f \in \mathcal{F} \mid \exists m \in M \exists r \in R \exists n \in N \text{ such that } f = \iota(m \cdot_M r, n) -_{\mathcal{F}} \iota(m, r \cdot_N n) \}.$$

$$H := \langle A_1 \cup A_2 \cup A_3 \rangle \leq (\mathcal{F}, +_{\mathcal{F}})$$

Since $\mathcal{F}$ is a $\mathbb{Z}$ -module, it follows that H is a $\mathbb{Z}$ -submodule of $\mathcal{F}$.

2.3 Step 3: Definition of $M \otimes_R N$

Define $M \otimes_R N := \mathcal{F}/H$. In full $\mathbb{Z}$ -module package, we have

$$(\mathbb{Z}, M \otimes_R N, +_3, \cdot_3)$$

For each $m \in M$ and $n \in N$, denote the coset of $\iota(m, n)$ in F/H by

$$m \otimes n := \iota(m, n) +_{\mathcal{F}} H.$$

These elements are called **simple tensors**.

2.4 Step 4: Defining relations in the quotient

Lemma 2.1. (*basic identities of $M \otimes_R N$*)

For all $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$, we have

$$\begin{aligned}(m_1 +_M m_2) \otimes n &= m_1 \otimes n +_3 m_2 \otimes n, \\ m \otimes (n_1 +_N n_2) &= m \otimes n_1 +_3 m \otimes n_2, \\ (m \cdot_M r) \otimes n &= m \otimes (r \cdot_N n).\end{aligned}$$

Proof. **First**, by definition of A_1 , the element

$$\begin{aligned}\iota(m_1 +_M m_2, n) -_{\mathcal{F}} \iota(m_1, n) -_{\mathcal{F}} \iota(m_2, n) &\in A_1 \subseteq H. \\ \implies \iota(m_1 +_M m_2, n) +_{\mathcal{F}} H &= (\iota(m_1, n) +_{\mathcal{F}} \iota(m_2, n)) +_{\mathcal{F}} H. \\ \implies (m_1 +_M m_2) \otimes n &= (\iota(m_1, n) +_{\mathcal{F}} H) +_3 (\iota(m_2, n) +_{\mathcal{F}} H) = m_1 \otimes n +_3 m_2 \otimes n\end{aligned}$$

Second, by definition of A_2 , the element

$$\begin{aligned}\iota(m, n_1 +_N n_2) -_{\mathcal{F}} \iota(m, n_1) -_{\mathcal{F}} \iota(m, n_2) &\in A_2 \subseteq H \\ \implies \iota(m, n_1 +_N n_2) +_{\mathcal{F}} H &= (\iota(m, n_1) +_{\mathcal{F}} \iota(m, n_2)) +_{\mathcal{F}} H \\ \implies m \otimes (n_1 +_N n_2) &= (\iota(m, n_1) +_{\mathcal{F}} H) +_3 (\iota(m, n_2) +_{\mathcal{F}} H) = m \otimes n_1 +_3 m \otimes n_2.\end{aligned}$$

Third, by definition of A_3 , the element

$$\begin{aligned}\iota(m \cdot_M r, n) -_{\mathcal{F}} \iota(m, r \cdot_N n) &\in A_3 \subseteq H \\ \implies \iota(m \cdot_M r, n) +_{\mathcal{F}} H &= \iota(m, r \cdot_N n) +_{\mathcal{F}} H \\ \implies (m \cdot_M r) \otimes n &= m \otimes (r \cdot_N n)\end{aligned}$$

□

2.5 Step 5: Some immediate additive consequences

Lemma 2.2. *For all $m \in M$ and $n \in N$, we have*

$$0_M \otimes n = 0_{M \otimes_R N} = m \otimes 0_N.$$

Proof. Using the first defining relation with $m_1 = m_2 = 0_M$, we obtain

$$0_M \otimes n = (0_M +_M 0_M) \otimes n = 0_M \otimes n +_3 0_M \otimes n \implies 0_{M \otimes_R N} = 0_M \otimes n$$

Similarly, using the second defining relation with $n_1 = n_2 = 0_N$, we obtain

$$m \otimes 0_N = m \otimes (0_N +_N 0_N) = m \otimes 0_N +_3 m \otimes 0_N \implies 0_{M \otimes_R N} = m \otimes 0_N$$

□

Lemma 2.3. *For all $m \in M$ and $n \in N$, we have*

$$(-m) \otimes n = -(m \otimes n)_{M \otimes_R N} = m \otimes (-n).$$

Proof. Observe that

$$0_M \otimes n = (m +_M (-m)) \otimes n = m \otimes n +_3 ((-m) \otimes n) \implies -(m \otimes n)_{M \otimes_R N} = (-m) \otimes n$$

Similarly,

$$m \otimes 0_N = m \otimes (n +_N (-n)) = m \otimes n +_3 m \otimes (-n) \implies -(m \otimes n)_{M \otimes_R N} = m \otimes (-n)$$

□

Now, take any $x \in M \otimes_R N$. Then there exists $f \in \mathcal{F}$ such that $x = f +_{\mathcal{F}} H$.

Since $f \in \mathcal{F} = F_{\mathbb{Z}}(M \times N)$, there exists a unique finitely supported coefficient function $c_f : \iota(M \times N) \rightarrow \mathbb{Z}$ s.t.

$$f = \sum_{\iota(m,n) \in \text{supp}(c_f)} c_f(\iota(m,n)) \cdot_{\mathcal{F}} \iota(m,n).$$

Equivalently, f is a finite $\mathbb{Z}$ -linear combination of basis elements of the form $\iota(m,n)$, so after expanding positive and negative integer coefficients, we may write

$$f = \sum_{\text{finite}} \pm \iota(m,n).$$

Therefore

$$x = f +_{\mathcal{F}} H = \left(\sum_{\text{finite}} \pm \iota(m,n) \right) +_{\mathcal{F}} H = \sum_{\text{finite}} m \otimes n.$$

Hence every element of $M \otimes_R N$ can be written as a finite sum of simple tensors.

Such an expression is generally not unique.

2.6 Step 6: Definition of a left S -module structure on $M \otimes_R N$

We now return to the question of endowing the abelian group $M \otimes_R N$ with a module structure.

Let R and S be rings with $1 \neq 0$. Suppose M is an (S, R) -bimodule, i.e.

- M is a left S -module,
- M is a right R -module,
- and for all $s \in S$, $r \in R$, and $m \in M$, we have

$$s \cdot_{(S,M)} (m \cdot_{(M,R)} r) = (s \cdot_{(S,M)} m) \cdot_{(M,R)} r$$

Further, let N be a left R -module.

For each fixed $s' \in S$, define a set map $\theta_{s'} : M \times N \rightarrow \mathcal{F}$ by,

$$(m, n) \mapsto \iota(s' \cdot_{(S,M)} m, n).$$

Since $\mathcal{F} = F_{\mathbb{Z}}(M \times N)$ is the free $\mathbb{Z}$ -module on the set $M \times N$, its universal property implies that there exists a unique group homomorphism

$$T_{s'} \in \text{Hom}_{\mathbb{Z}}(\mathcal{F}, \mathcal{F}) \quad \text{s.t.} \quad \theta_{s'} = T_{s'} \circ \iota$$

For illustration, if

$$f = \sum_{i=1}^k z_i \cdot_{\mathcal{F}} \iota(m_i, n_i) \in \mathcal{F},$$

then

$$T_{s'}(f) = \sum_{i=1}^k z_i \cdot_{\mathcal{F}} \iota(s' \cdot_{(S,M)} m_i, n_i).$$

Lemma 2.4. For every $s' \in S$, we have

$$T_{s'}(H) \subseteq H.$$

Proof. Since $H = \langle A_1 \cup A_2 \cup A_3 \rangle$, it suffices to show that $T_{s'}$ sends $A_1 \cup A_2 \cup A_3$ into H .

First, let

$$g = \iota(m_1 +_M m_2, n) -_{\mathcal{F}} \iota(m_1, n) -_{\mathcal{F}} \iota(m_2, n) \in A_1.$$

$$\begin{aligned} \implies T_{s'}(g) &= \iota(s' \cdot_{(S,M)} (m_1 +_M m_2), n) -_{\mathcal{F}} \iota(s' \cdot_{(S,M)} m_1, n) -_{\mathcal{F}} \iota(s' \cdot_{(S,M)} m_2, n) \\ &= \iota((s' \cdot_{(S,M)} m_1) +_M (s' \cdot_{(S,M)} m_2), n) -_{\mathcal{F}} \iota(s' \cdot_{(S,M)} m_1, n) -_{\mathcal{F}} \iota(s' \cdot_{(S,M)} m_2, n). \end{aligned}$$

Since M is a left S -module, left multiplication by s' is additive on M . Hence $T_{s'}(g) \in A_1 \subseteq H$.

Second, let

$$g = \iota(m, n_1 +_N n_2) -_{\mathcal{F}} \iota(m, n_1) -_{\mathcal{F}} \iota(m, n_2) \in A_2.$$

$$\implies T_{s'}(g) = \iota(s' \cdot_{(S,M)} m, n_1 +_N n_2) -_{\mathcal{F}} \iota(s' \cdot_{(S,M)} m, n_1) -_{\mathcal{F}} \iota(s' \cdot_{(S,M)} m, n_2) \in A_2 \subseteq H.$$

Third, let

$$g = \iota(m \cdot_{(M,R)} r, n) -_{\mathcal{F}} \iota(m, r \cdot_N n) \in A_3.$$

$$\begin{aligned} \implies T_{s'}(g) &= \iota(s' \cdot_{(S,M)} (m \cdot_{(M,R)} r), n) -_{\mathcal{F}} \iota(s' \cdot_{(S,M)} m, r \cdot_N n) \\ &= \iota((s' \cdot_{(S,M)} m) \cdot_{(M,R)} r, n) -_{\mathcal{F}} \iota(s' \cdot_{(S,M)} m, r \cdot_N n). \end{aligned}$$

Hence $T_{s'}(g) \in A_3 \subseteq H$. □

Let $\pi : \mathcal{F} \rightarrow \mathcal{F}/H = M \otimes_R N$ be the natural quotient map.

Since $T_{s'}(H) \subseteq H$, it follows that

$$H \subseteq \ker(\pi \circ T_{s'}).$$

Hence $\pi \circ T_{s'}$ factors through $\mathcal{F}/H$. Thus there exists a group homomorphism

$$\bar{T}_{s'} \in \text{Hom}_{\mathbb{Z}}(M \otimes_R N, M \otimes_R N) \quad \text{s.t.} \quad \bar{T}_{s'}(f +_{\mathcal{F}} H) = T_{s'}(f) +_{\mathcal{F}} H \quad \forall f \in \mathcal{F}.$$

For illustration, if

$$x = \sum_{i=1}^k m_i \otimes n_i \in M \otimes_R N,$$

then

$$\bar{T}_{s'}(x) = \sum_{i=1}^k \bar{T}_{s'}(m_i \otimes n_i) = \sum_{i=1}^k \iota(s' \cdot_{(S,M)} m_i, n_i) +_{\mathcal{F}} H = \sum_{i=1}^k (s' \cdot_{(S,M)} m_i) \otimes n_i.$$

We now define $\diamond : S \times (M \otimes_R N) \rightarrow M \otimes_R N$ as,

$$(s', x) \mapsto \bar{T}_{s'}(x)$$

Theorem 2.1. *With the action $\diamond$ defined above, $M \otimes_R N$ is a left S -module.*

Proof. We verify the left S -module axioms.

(identity) Let $x \in M \otimes_R N$. Then

$$1_S \diamond x = \bar{T}_{1_S}(x) = x,$$

since for every simple tensor $m \otimes n$ we have

$$1_S \diamond (m \otimes n) = (1_S \cdot_{(S,M)} m) \otimes n = m \otimes n.$$

(additivity in the scalar) Let $s_1, s_2 \in S$ and let $x = \sum_{i=1}^k m_i \otimes n_i \in M \otimes_R N$, we have

$$\begin{aligned} (s_1 +_S s_2) \diamond x &= \sum_{i=1}^k ((s_1 +_S s_2) \cdot_{(S,M)} m_i) \otimes n_i \\ &= \sum_{i=1}^k ((s_1 \cdot_{(S,M)} m_i) +_M (s_2 \cdot_{(S,M)} m_i)) \otimes n_i \\ &= \sum_{i=1}^k (s_1 \cdot_{(S,M)} m_i) \otimes n_i +_3 \sum_{i=1}^k (s_2 \cdot_{(S,M)} m_i) \otimes n_i \\ &= s_1 \diamond x +_3 s_2 \diamond x. \end{aligned}$$

(additivity in the tensor variable) Let $s \in S$ and $x, y \in M \otimes_R N$, we have

$$s \diamond (x +_3 y) = \bar{T}_s(x +_3 y) = \bar{T}_s(x) +_3 \bar{T}_s(y) = s \diamond x +_3 s \diamond y.$$

(associativity of scalar multiplication) Let $s_1, s_2 \in S$ and let $x = \sum_{i=1}^k m_i \otimes n_i$, we have

$$\begin{aligned}
 s_1 \diamond (s_2 \diamond x) &= s_1 \diamond \left(\sum_{i=1}^k (s_2 \cdot_{(S,M)} m_i) \otimes n_i \right) \\
 &= \sum_{i=1}^k (s_1 \cdot_{(S,M)} (s_2 \cdot_{(S,M)} m_i)) \otimes n_i \\
 &= \sum_{i=1}^k ((s_1 \cdot_S s_2) \cdot_{(S,M)} m_i) \otimes n_i \\
 &= (s_1 \cdot_S s_2) \diamond x.
 \end{aligned}$$

Therefore $M \otimes_R N$ is a left S -module. □

2.6.1 Right S -module structure on $M \otimes_R N$, and an Alternative Universal Property Construction

Theorem 2.2. (*alternative construction, right S -module on $M \otimes_R N$*)

Let M be a right R -module and let N be an (R, S) -bimodule.

Then, $M \otimes_R N$ has the structure of a right S -module, where for every $x \in M \otimes_R N$ and $s \in S$, the scalar multiplication is defined by

$$x \diamond s := \rho_s(x),$$

where $\rho_s \in \text{Hom}_{\mathbb{Z}}(M \otimes_R N, M \otimes_R N)$ is the unique group homomorphism satisfying

$$\rho_s(m \otimes n) = m \otimes (n \cdot_{(N,S)} s) \quad \forall m \in M, \forall n \in N.$$

For illustration, for every finite sum of simple tensors, we have

$$\left(\sum_{\text{finite}} m_i \otimes n_i \right) \diamond s = \sum_{\text{finite}} m_i \otimes (n_i \cdot_{(N,S)} s).$$

Proof. Fix $s \in S$. Define a map $\varphi_s : M \times N \rightarrow M \otimes_R N$ by,

$$\varphi_s(m, n) := m \otimes (n \cdot_{(N,S)} s)$$

We claim that φ_s is R -balanced.

First, let $m_1, m_2 \in M$ and $n \in N$. Then

$$\begin{aligned} \varphi_s(m_1 +_M m_2, n) &= (m_1 +_M m_2) \otimes (n \cdot_{(N,S)} s) \\ &= m_1 \otimes (n \cdot_{(N,S)} s) +_3 m_2 \otimes (n \cdot_{(N,S)} s) \\ &= \varphi_s(m_1, n) +_3 \varphi_s(m_2, n). \end{aligned}$$

Second, let $m \in M$ and $n_1, n_2 \in N$. Then

$$\begin{aligned} \varphi_s(m, n_1 +_N n_2) &= m \otimes ((n_1 +_N n_2) \cdot_{(N,S)} s) \\ &= m \otimes ((n_1 \cdot_{(N,S)} s) +_N (n_2 \cdot_{(N,S)} s)) \\ &= m \otimes (n_1 \cdot_{(N,S)} s) +_3 m \otimes (n_2 \cdot_{(N,S)} s) \\ &= \varphi_s(m, n_1) +_3 \varphi_s(m, n_2). \end{aligned}$$

Third, let $m \in M$, $r \in R$, and $n \in N$. Then

$$\begin{aligned} \varphi_s(m \cdot_M r, n) &= (m \cdot_M r) \otimes (n \cdot_{(N,S)} s) \\ &= m \otimes (r \cdot_{(R,N)} (n \cdot_{(N,S)} s)) \\ &= m \otimes ((r \cdot_{(R,N)} n) \cdot_{(N,S)} s) \\ &= \varphi_s(m, r \cdot_{(R,N)} n). \end{aligned}$$

Here, in the third equality, we used the bimodule compatibility on N .

Thus φ_s is R -balanced. By Theorem 10, $\exists! \rho_s \in \text{Hom}_{\mathbb{Z}}(M \otimes_R N, M \otimes_R N)$ s.t.

$$\varphi_s = \rho_s \circ \iota.$$

Equivalently,

$$\rho_s(m \otimes n) = m \otimes (n \cdot_{(N,S)} s) \quad \forall m \in M, \forall n \in N.$$

Now define $\diamond : (M \otimes_R N) \times S \rightarrow M \otimes_R N$ as,

$$(x, s) \mapsto \rho_s(x)$$

Since each ρ_s is a group homomorphism, for every finite sum of simple tensors we have

$$\left(\sum_{\text{finite}} m_i \otimes n_i \right) \diamond s = \rho_s \left(\sum_{\text{finite}} m_i \otimes n_i \right) = \sum_{\text{finite}} \rho_s(m_i \otimes n_i) = \sum_{\text{finite}} m_i \otimes (n_i \cdot_{(N,S)} s).$$

Hence the displayed formula is well-defined.

It remains to verify the right S -module axioms.

(identity) Let $x = \sum_{\text{finite}} m_i \otimes n_i \in M \otimes_R N$. Then

$$\begin{aligned} x \diamond 1_S &= \sum_{\text{finite}} m_i \otimes (n_i \cdot_{(N,S)} 1_S) \\ &= \sum_{\text{finite}} m_i \otimes n_i \\ &= x. \end{aligned}$$

(additivity in the tensor variable) Let $x, y \in M \otimes_R N$ and let $s \in S$. Since ρ_s is a group homomorphism, we have

$$(x +_3 y) \diamond s = \rho_s(x +_3 y) = \rho_s(x) +_3 \rho_s(y) = x \diamond s +_3 y \diamond s.$$

(additivity in the scalar) Let $x = \sum_{\text{finite}} m_i \otimes n_i \in M \otimes_R N$ and let $s_1, s_2 \in S$. Then

$$\begin{aligned} x \diamond (s_1 +_S s_2) &= \sum_{\text{finite}} m_i \otimes (n_i \cdot_{(N,S)} (s_1 +_S s_2)) \\ &= \sum_{\text{finite}} m_i \otimes ((n_i \cdot_{(N,S)} s_1) +_N (n_i \cdot_{(N,S)} s_2)) \\ &= \sum_{\text{finite}} m_i \otimes (n_i \cdot_{(N,S)} s_1) +_3 \sum_{\text{finite}} m_i \otimes (n_i \cdot_{(N,S)} s_2) \\ &= x \diamond s_1 +_3 x \diamond s_2. \end{aligned}$$

(associativity of scalar multiplication) Let $x = \sum_{\text{finite}} m_i \otimes n_i \in M \otimes_R N$ and let $s_1, s_2 \in S$. Then

$$\begin{aligned} (x \diamond s_1) \diamond s_2 &= \left(\sum_{\text{finite}} m_i \otimes (n_i \cdot_{(N,S)} s_1) \right) \diamond s_2 \\ &= \sum_{\text{finite}} m_i \otimes ((n_i \cdot_{(N,S)} s_1) \cdot_{(N,S)} s_2) \\ &= \sum_{\text{finite}} m_i \otimes (n_i \cdot_{(N,S)} (s_1 \cdot_S s_2)) \\ &= x \diamond (s_1 \cdot_S s_2). \end{aligned}$$

Therefore $M \otimes_R N$ is a right S -module. □